ABB Welcome ESP32 Bus Interface – Quick Start Guide

2026-11-12 · Firmware 1.1.0 · For ABB Welcome and Busch-Welcome 2-wire systems

Before you start

This board connects your ABB Welcome or Busch-Welcome 2-wire intercom to Home Assistant over Wi-Fi. It is pre-flashed, bus-powered and needs no cloud account. Four steps get it running: wire it, join it to Wi-Fi, add it to Home Assistant, then match it to your installation.

In the box

- 1× assembled ESP32 interface board with ESPHome firmware 1.1.0 pre-installed
- This guide
- The firmware configuration, published at github.com/VVlasy/abb-doorbell-esphome. Nothing to copy by hand: Step 4 pulls it in automatically.

You need

- An ABB Welcome or Busch-Welcome 2-wire system with 20–30 V DC on the bus
- A 2.4 GHz Wi-Fi network (the ESP32 cannot see 5 GHz networks)
- Home Assistant with the ESPHome integration, plus the ESPHome Device Builder add-on for Step 4
- A small flat screwdriver, a multimeter and a phone or laptop

Your unit

Device name (also the name of its setup hotspot)	abb-doorbell-_____
Shown in Home Assistant as	ABB Doorbell _____
Firmware	ESPHome, project victor.abb_doorbell 1.1.0

The six characters are the last digits of the board's Wi-Fi MAC address and are unique to your unit. You will see them in the name of the hotspot the board opens in Step 2; write them in above for later reference.

Safety

- Low voltage only. Connect the board to the 2-wire intercom bus and nothing else. Never connect it to mains.
- Measure the bus voltage before connecting. It must be 20–30 V DC; the board tolerates 36 V DC at most.
- Switch off the intercom power supply or system controller while you wire the board.
- The board is supplied without an enclosure. Mount it where nothing metallic can touch it and keep the antenna end out of metal boxes.
- Works with 2-wire ABB Welcome and Busch-Welcome systems only, not with ABB Welcome IP. It carries no audio or video and, according to the project page, may slightly reduce audio or video quality on some systems. This is not an official ABB product.

Step 1 – Wire the board to the bus

Connect the board in parallel with your indoor station, on the same two bus wires that already feed it. Nothing else in the intercom installation changes, and the board takes its power from the bus.

1. Find the two bus wires at your indoor station (terminals usually marked **a** and **b**) or in the flush-mounting box behind it.
2. With the intercom powered, measure the DC voltage between the two wires. Expect 20–30 V DC and note which wire is positive. If you read more than 36 V, AC, or nothing, stop and contact support.
3. Switch off the intercom power supply or system controller.

4. Connect the positive wire to **BUS+ (A)** and the negative wire to **BUS− (B)** on the green screw terminal. Double up on the indoor station's terminals or use a junction box; no bare strands may stick out.
5. Place the board in a dry spot where nothing metallic can touch it, with the antenna end away from metal. A flush-mounting box behind or next to the indoor station usually has room.
6. Switch the power back on. **PWR** lights, **RX** flickers whenever there is traffic on the bus, and **WIFI** blinks slowly until the board is online.

Reversed wires do no harm: the board has reverse-polarity protection and a resettable fuse. If PWR stays dark, swap the two wires. If **TX** stays lit continuously, disconnect the board and contact support.

Marking on the board	Meaning
BUS+ (A) / BUS− (B)	2-wire bus connection, 20–30 V DC
PWR	Board has power
RX	Message received from the bus
TX	Board is sending on the bus (short flash when you unlock or stop the bell)
WIFI	Status LED: slow blink = not connected yet, off = connected to Wi-Fi and Home Assistant, fast blink = error
RESET button	Restarts the board; settings are kept
Serial header DTR / TX / RX / 3V3 / RTS / GND	For advanced users only: 3.3 V USB-serial flashing and Wi-Fi setup via web.esphome.io

Step 2 – Connect the board to your Wi-Fi

The board ships without Wi-Fi credentials, so it opens its own hotspot as soon as it has power. You hand it your network details from a phone or laptop; this takes about two minutes.

1. On your phone or laptop, open the Wi-Fi list and join the open network **abb-doorbell-xxxxxx** (the device name of your unit, see *Your unit* above). It has no password.
2. A setup page opens by itself. If nothing appears within a few seconds, open a browser and go to **http://192.168.4.1**.
3. Pick your home network from the list, type its password and save.
4. The board restarts and joins your network, and the hotspot disappears. **WIFI** keeps blinking slowly until Home Assistant connects in Step 3.

Good to know

- The ESP32 supports 2.4 GHz only. A network that uses the same name on both bands is fine; a 5 GHz-only network is not.
- Use the network your Home Assistant server is on. Guest networks and networks with client isolation stop Home Assistant from finding the board.
- Wrong password? After one to two minutes of failed attempts the hotspot comes back. Join it again and re-enter the details.
- Give the board a fixed IP address with a DHCP reservation in your router. Step 4 and any later troubleshooting get easier.
- New router or new Wi-Fi password later? The hotspot reappears by itself within about two minutes; repeat the steps above.
- Advanced alternative: with a 3.3 V USB-serial adapter on the serial header you can also set Wi-Fi from web.esphome.io (Improv Serial).

Step 3 – Add the board to Home Assistant

Home Assistant finds the board on its own within about a minute of it joining your Wi-Fi.

1. Open **Settings** → **Devices & services**. A *Discovered* card **ABB Doorbell xxxxxx** (ESPHome) is waiting. Click **Add**, then **Submit**.
2. Pick an area such as *Hallway* and finish. The device and the entities listed in the next section are now in Home Assistant, and **WIFI** stops blinking.

Not discovered? Click **Add integration**, choose **ESPHome** and enter the board's IP address (from your router's client list, hostname abb-doorbell-xxxxxx) or `abb-doorbell-xxxxxx.local`, port 6053. Discovery needs Home Assistant and the board on the same network; separate VLANs need an mDNS repeater or this manual route.

Encryption key: the board ships without an API encryption key, so Home Assistant does not ask for one now. The link is unencrypted until Step 4, where adopting the board adds a key automatically.

The board's own web page: open `http://abb-doorbell-xxxxxx.local` (or the IP address) in a browser. It lists the same entities, lets you press the buttons and shows a live log of bus messages. It works without Home Assistant and is handy in Step 4.

What you get in Home Assistant

The board appears as one device with the entities below. Entity IDs follow the pattern `binary_sensor.abb_doorbell_xxxxxx_doorbell_outdoor`.

Important: the firmware ships with placeholder bus addresses and a placeholder door-opener code. Until you finish Step 4, *Front Door* and *Stop Doorbell* do nothing, and the doorbell sensors may stay silent.

Entity	Type	What it does
Doorbell outdoor	Binary sensor	On for about 1 s when someone rings at the outdoor station (building entrance). Use it as an automation trigger.
Doorbell indoor	Binary sensor	On for about 1 s when the bell button wired directly to your indoor station is pressed, typically the one at your apartment door.
Front Door	Lock	*Unlock* sends the door-opener command to the bus. Shows *Unlocking* for 3 s when the door confirms, then returns to *Locked*. It does not sense the real door; *Locked* is simply its resting state.
Stop Doorbell	Button	Ends the call and silences the ringtone on your indoor station.
Doorbell Intercom	Sensor (text)	The last message seen on the bus, decoded. You use it in Step 4 to find your addresses.
Internal Temperature	Sensor	Temperature of the ESP32 chip, for diagnostics.
Reboot, Reboot (Safe Mode)	Buttons	Restart the board. Safe mode starts only Wi-Fi and the update function, for recovery.

Typical automations: a phone notification (with a camera snapshot) when *Doorbell outdoor* turns on, a wall button that unlocks the door and presses *Stop Doorbell*, or the hallway light switching on when the bell rings.

Step 4 – Match the firmware to your installation

Every ABB Welcome installation uses its own bus addresses and its own door-opener code. The firmware ships with placeholders (indoor station `0x1001`, outdoor station `0x2001`, door `0x4001`, code `AB CD EF`) that you replace with the values seen on your bus. Plan 30–60 minutes the first time; the full background is in Mat931's customisation guide.

4a. Record what your bus says

The *Doorbell Intercom* sensor shows the last message decoded from the bus; the board's web page shows every message in its log. Note the exact time of each action below and the lines that appear within 1–2 seconds of it.

1. Ring the bell at the outdoor station.
2. Press the door-opener button on your indoor station.
3. If a bell button at your apartment door is wired to the indoor station, press it.
4. Start a call to the outdoor station from the indoor station without anyone ringing. Messages that also appear now are not doorbell messages; ignore them.

A message looks like this:

```
55.FF.… [1001 > 4001] Type: 0D, Data: AB.CD.EF
```

Inside the brackets stand the source and destination addresses, then the message type and the data bytes. A `»` instead of `>` marks a retransmission. Types `80` and above are replies.

4b. Work out your values

What you need	How to recognise it in the log
Address length	Addresses with 4 characters: 2-byte system, keep `three_byte_address: false`. Addresses with 6 characters: 3-byte system, set `three_byte_address: true`.
Indoor station address	The address that appears most often, typically `10xx` or `0001xx`. The last two characters are your apartment number in hexadecimal (apartment 12 shows as `0c`).
Outdoor station address	Source of the `Type: 01` message that arrives when the bell rings; its destination is your indoor station.
Door address and opener code	When you press the door opener: a `Type: 0D` (sometimes `0E`) message from your indoor station. Its destination is the door address and its `Data` bytes are the code. A `Type: 8D` (or `8E`) reply from the door confirms it.
Indoor doorbell	A `Type: 11` message with your indoor station as source.

Only ever send messages you have seen your own indoor station send. The bus is shared with your neighbours, and a guessed command can ring or unlock the wrong door.

4c. Adopt the board in the ESPHome Device Builder

Adopting creates a small configuration of your own that pulls the published firmware in as a package. You only ever edit the substitutions in it.

1. Install the **ESPHome Device Builder** add-on (**Settings** → **Add-ons** → **Add-on store**) and open it.
2. The board is listed at the top as discovered. Click **Adopt**, enter your Wi-Fi name and password if asked, then **Install** when offered. This first build takes 5–15 minutes and adds an API encryption key. If Home Assistant asks for that key afterwards, copy it from the `api:` section of the new configuration.
3. Click **Edit** on the board's card and add your values from 4b under `substitutions:`, keeping the quotes. The two name lines are already there.

4. **Save**, then **Install → Wirelessly**. The board restarts with your values.
5. Test: ring the bell and watch *Doorbell outdoor* turn on, then unlock *Front Door* and listen for the door release.

Example for a system with 3-byte addresses and an `0E`-type opener:

```
substitutions:
  name: abb-doorbell-xxxxxx           # filled in by Adopt
  friendly_name: ABB Doorbell xxxxxx  # filled in by Adopt
  indoor_station_address: "0x000110"
  outdoor_station_address: "0x000201"
  door_address: "0x000401"
  three_byte_address: "true"
  door_open_type: "0x0e"
  door_open_reply_type: "0x8e"
  door_open_code: "0x00, 0x00, 0x00"
  unlock_time: "3s"
```

All substitutions and their meaning are listed at github.com/VVlasy/abb-doorbell-esphome. Your Wi-Fi details live in the adopted configuration, so the board no longer depends on what you entered in Step 2.

Without the add-on: copy `abb-doorbell.yaml` from that page into any ESPHome dashboard, set `name:` to your unit's full device name, delete the line `name_add_mac_suffix: true`, edit the substitutions and install wirelessly. If the dashboard cannot reach the board, add `use_address: 192.168.x.x` with the board's IP address under `wifi:`.

Troubleshooting

The **WIFI** LED tells you most of what you need: slow blinking means the board is not yet connected to Wi-Fi or to Home Assistant, off means everything is connected, fast blinking means an internal error.

Symptom	Likely cause	What to do
PWR stays dark	No bus voltage, or wires reversed	Measure the bus, swap the two wires, check the screw terminals.
Hotspot abb-doorbell-xxxxxx never appears	No power, or the board is already connected to a Wi-Fi network	Check PWR; look for the board in your router's client list. The hotspot only opens while the board has no working Wi-Fi.
WIFI blinks slowly for more than two minutes after Step 2	Wrong password, 5 GHz-only network, or Home Assistant not yet connected	Wait for the hotspot to return and re-enter the details. After Step 3 the LED goes off.
WIFI blinks fast	Internal error	Press RESET. If it persists, contact support.
Not discovered in Home Assistant	Different network or VLAN, mDNS blocked	Add it manually by IP address (Step 3).
No *Adopt* button in the ESPHome Device Builder	Board not on the same network as Home Assistant, or mDNS blocked	Fix the network (Step 3) or use the manual route at the end of Step 4.
Shows *unavailable* for a moment every 15 minutes	Home Assistant not connected	Normal before Step 3: without a connected Home Assistant the board restarts every 15 minutes (an ESPHome default).
Doorbell sensors never turn on	Bus addresses do not match your installation	Step 4. If RX does not flicker when the bell rings, the board is not on the bus: check the wiring.
Log shows the outdoor station's messages but never your own indoor station's	The indoor station's bus-termination switch (often marked RC or *terminating resistor*) is on	Switch it off if your installation allows it, then watch the log again.
Front Door unlock does nothing	Placeholder door code or wrong address, or a 3-byte system	Step 4: check `door_address`, `door_open_code`, `door_open_type` (`0x0d` or `0x0e`) and `three_byte_address`.
Stop Doorbell does not silence the ringtone	Outdoor station address wrong	Step 4.
TX stays lit continuously	Firmware or hardware fault	Disconnect the board from the bus and contact support.
Device Builder cannot reach the board for a wireless install	Name mismatch, only when you pasted the file instead of adopting	Use the full name `abb-doorbell-xxxxxx`, delete `name_add_mac_suffix: true`, or add `use_address` with the board's IP.
Board offline after a router or password change	Old Wi-Fi credentials	The hotspot returns after about two minutes; repeat Step 2.
Board keeps restarting after a firmware change	Faulty configuration	After ten failed boots it enters safe mode (Wi-Fi and update only); install a corrected configuration wirelessly.

There is no factory-reset button. To change Wi-Fi, use the hotspot that appears when the old network is gone; to change anything else, repeat Step 4.

Safety, licence and support

Safety and liability. The board is for the low-voltage 2-wire intercom bus only and must be kept dry and clear of metal. You are responsible for the wiring and for the configuration you install; the seller is not liable for damage caused by incorrect wiring, over-voltage or modified firmware. In an apartment building the bus is shared with your neighbours: the board only listens and sends the same commands your indoor station sends, but check your house rules before connecting anything to a shared system.

Open-source hardware. The board is an independently assembled build of the *esp32-doorbell-bus-interface* design, copyright 2024 Mat931, released under the CERN Open Hardware Licence v2 Weakly Reciprocal (CERN-OHL-W v2). A share of the proceeds from every board goes to the original author. The firmware is ESPHome, open source under its own licences. This is not an official ABB product; ABB, ABB Welcome and Busch-Welcome are trademarks of their owners and are used only to describe compatibility.

Support. Contact the seller through eBay messages. Include your order number, what you tried, and if possible a few lines from the *Doorbell Intercom* sensor or the board's log. Firmware and bus questions are also discussed in the project's GitHub Discussions.

Further reading

- Firmware configuration and substitution reference: github.com/VVlasy/abb-doorbell-esphome
- Hardware project, design files and documentation: github.com/Mat931/esp32-doorbell-bus-interface
- Finding your addresses and door code: how_to_customize.md
- Serial flashing and bus wiring notes: how_to_flash.md
- Example configuration on the ESPHome device list: devices.esphome.io
- ESPHome in Home Assistant: home-assistant.io/integrations/esphome
- Wi-Fi hotspot and captive portal: esphome.io/components/captive_portal
- Status LED: esphome.io/components/status_led